

Model Comparison Assignment

Decisions under uncertainty in a cable car system

Part A - Environment description

This model examines the behavior of a passenger who faces a repeated decision in a cable car system. The passenger is alone on the platform, i.e., there is no line to enter a cable car. At the arrival of a cable car, the passenger sees if the car is empty or occupied (each occurring with a specified probability), and may either board the car immediately, regardless if it is empty or not, in order to save time, or the agent can choose to board a car only if it is empty in order to ensure a better travel experience.

Upon the agent's arrival at the cable car platform, the passenger has two discrete choices:

1. **Safe choice (S):** Board the first cable car to arrive at the platform.
2. **Risky choice (R):** Board the first *empty* car that arrives at the platform.

The decision involves a tradeoff between two factors: time cost and ride comfort. Riding in an occupied car results in a varying, random degree of discomfort, whereas riding alone maximizes the enjoyment from the ride. However, spending time in order to wait for an empty car harms the agent's utility. Although both choices possess randomness, riskiness in this context is with regards to waiting time: Safe is guaranteed to not wait at all, whereas Risk has an unknown waiting time.

The agent is familiar with only their own experience (their decisions and respective outcomes); they don't know the forgone outcome. If the agent chooses Safe, they receive the resulting utility but gain no information about what would have occurred had they chosen Risk because they don't know how long they would have needed to wait. If the agent chose Risk, although they see if the first car to arrive is occupied, they do not know how much disutility would have been incurred from riding with that specific passenger. This restricts learning to actual experiences and eliminates counterfactual regret. It is worth noting that in the case when the first car to arrive is empty, Risk and Safe imply the same action (to board the car) and lead to the same utility.

The following values and parameters are defined:

- ❖ Fixed utility received from riding an empty car = **20**
- ❖ $D \sim N(3, 1^2)$: Disutility experienced when riding in an occupied car. This is a random variable because sometimes the passenger you ride with isn't so awful, and other times the other passenger is talking loudly on the phone, eating smelly food, etc.
- ❖ $p = 0.15$: the probability a car is empty. $(1-p)$ is the probability a car is occupied.
- ❖ $T \sim \text{Geo}(0.15)$: The T^{th} car is the first empty car to arrive, which means $(T-1)$ is the number of cars skipped because they are occupied.
- ❖ $f(T-1)$: When risk is chosen, this function represents the cost of waiting. f is a convex and negative function. The marginal cost of a delay is increasing, i.e., small delays are tolerable, whereas large delays significantly harm utility.

We chose the following parameters, attempting to quantify our experience riding the cable cars:

Choice	Utility	Expected Utility
Safe	$U_{\text{safe}} = \begin{cases} 20 & \text{with probability } 0.15 \\ -D & \text{with probability } 0.85 \end{cases} \quad D \sim N(3, 1^2)$	$E[U_S] = 0.45^*$
Risk	$U_R = 20 - 0.285(T-1)^2$	$E[U_R] = 0.086^{**}$

$$^* E[U_{\text{safe}}] = 0.15 \cdot 20 + (1 - 0.15) \cdot E[D] = 0.45$$

$$^{**} E[U_{\text{risk}}] = E[20 - 0.285(T-1)^2] = 20 - 0.285 \cdot E[(T-1)^2] = 20 - 0.285 \cdot \frac{(1-0.15)(2-0.15)}{0.15^2} = 0.086$$

Given the chosen parameters and comparing expected utilities, **Safe is the optimal choice.**

Part B - Model Descriptions

Agent 1: Explorative

This agent is a version of running average model. Safe or Risk is chosen randomly in the first trial. The agent continues with that choice, but with a certain probability chooses the other option. After both choices have been chosen at least once, the agent chooses the option with the higher average from past experiences yet still has a probability of choosing the option with the lower average. The probability of choosing the worse option decreases over time. This agent represents someone who rationally chooses according to the average experience but keeps an open mind to explore other options sometimes, especially in the first trials.

Behavior prediction: Because Safe has the higher expected value (EV), this agent will learn to choose Safe more than Risk; however, the agent will not quite get to 0% Risk rate because of her explorative nature.

Agent 2: Threshold

This agent is a version of the Win-Stay-Lose-Shift model. Losing in this context means that the agent received negative utility from their choice. The agent chooses a Risk or Safe randomly in the first choice and then continues choosing an option so long as the option's previous outcome was not less than the threshold of 0. Should the outcome fall below 0, the agent switches strategies. This agent's behavior represents someone who impulsively changes their choice after a bad experience, not considering the overall average experience from either option.

Behavior prediction: The Risk rate for this agent will be roughly 85%. The agent switches her choice after receiving negative utility. In 85% of the trials, the Safe option gives D , which is almost always negative given its distribution. Likewise, the probability of a negative outcome from Risk is about 23%.

Agent 3: Small Recent Sample

This agent chooses between Risk and Safe according to what will provide a better experience most of the time, disregarding the full history of experiences and the average experience from either choice. The agent chooses randomly for the first $K=4$ choices, and then chooses the option that provided the best average outcome in the most recent $K=4$ trials.

Behavior prediction: After running a simulation, the probability of Risk giving a greater utility than Safe is approximately 0.667. However, the EV of Safe is higher than that of Risk. So, in a small sample, Risk will be better most of the time, but average of Risk in the small sample won't necessarily be great than the Safe average. Accordingly, the Risk rate will be less than 0.667, though unclear by how much.

Agent 4: Maximin

This agent is pessimistic, assumes the worst-case scenario from each choice will happen, and therefore, chooses the option with the highest received minimum outcome. The agent tries both Risk and Safe at first, and then tracks the worst outcome from either option throughout all trials, and chooses the option with the better worst outcome.

Behavior prediction: Given that $D \sim N(3, 1^2)$, $-D$ almost always falls between 0 and -6. Given that $T \sim \text{Geo}(p=0.15)$, the chance of getting an outcome less than -6 is about 19.7%. In other words, Risk will likely give the lowest worst outcome, and the Risk rate will eventually drop to 0%.

Overall reward prediction between agents:

We believe that the accumulated utility at the end of trials will be determined by the Risk rate: the higher the risk rate, the lower the cumulative rewards, since the EV of Safe is higher than that of risk. Accordingly, Maximin will have the highest accumulated utility, then Explorative, then Small Recent Sample, and lastly, Threshold.

Part D – Comparisons and Conclusions

Comparison of agent results:

The simulation revealed that the best performers, from best to worst, are: Maximin, Explorative, Small Recent Sample (SRS), and lastly, Threshold. Interpreting these results, the best strategy in this setting was employed by the “pessimist” who considers worst-case scenarios; conversely, the worst strategy was employed by the “impulsive” agent who reacts unwisely to negative utility.

With the above being said, the true indicator of performance was the choice rate of Risk, as seen in *Figure 2*. The order of the agents' risk rate, from highest to lowest is Threshold, SRS, Explorative, and Maximin - the exact inverse of the performance ranking. We can induce that the higher the risk rate, the lower the cumulative reward received, as predicted in Part B. *Figure 2* also shows that all agents but Explorative had quick convergences; Explorative learned more slowly, likely due to her explorative nature.

Prediction VS Results:

Maximin: As predicted, the risk rate of this agent dropped to 0%, and it did so quickly, resembling the hot stove effect. The agent early in the simulation experienced from Risk a bad outcome that outweighed the worst outcome from Safe, making the agent avoid Risk the rest of the simulation.

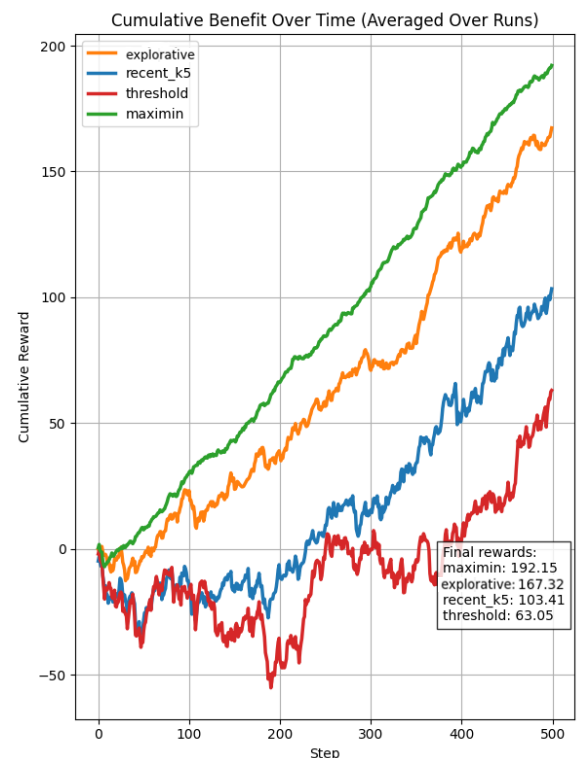
Explorative: The agent indeed learned to choose Safe more often than Risk as predicted; however, we were unable to predict to which Risk rate this agent would converge. The simulation shows about an 18% Risk rate (see *Figure 2*).

SSR: The Risk rate indeed proved to be less than 0.667, converging at about 50%. We can induce that the higher frequency of Risk being more beneficial than Safe was offset by the higher EV of the Safe option, leading the agent to choose either option 50% of the time (see *Figure 2*).

Threshold: The Risk rate for this agent was predicted to be roughly 85%, while in practice, the Risk rate was slightly lower, converging to approximately 80% (see *Figure 2*).

Among our main insights is the significance of following the expected value. Those who chose Safe, which had the highest EV, most often achieved the greatest utility. Surprisingly, Maximin performed best, despite that in real life, making a decision based on the worst-case scenario often leads to suboptimal choices. Explorative, being a version of the running average model, understandably performed well. Conversely, Threshold is a version of Win-Stay-Lose-Shift, which is a private case of SRS, so predictably they'd both perform worse than the other models due to their focus on only recent outcomes.

Figure 1: Agents' Cumulative Utility Over Time



Percentage of 'R' Actions per Step (Each Agent)

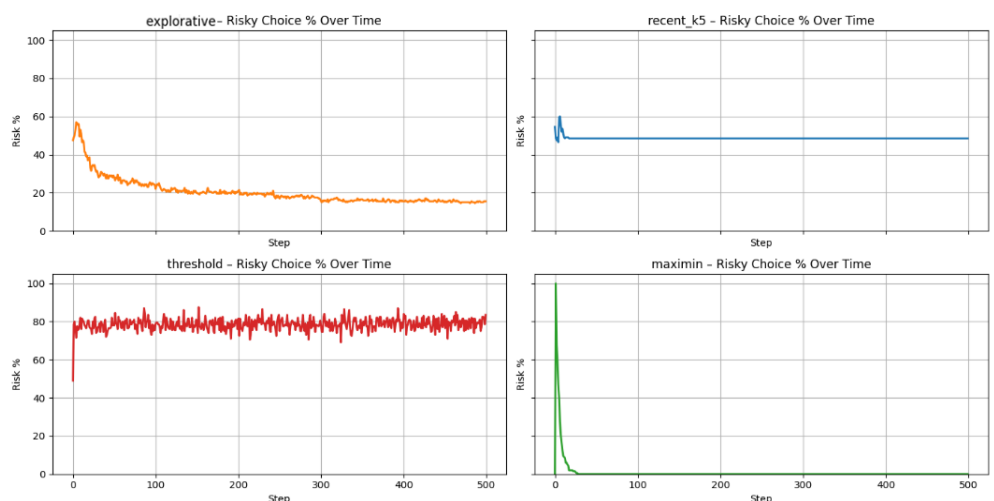


Figure 2: Risk Rate Over Time